

PRITHVIRAJ MODY

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EDUCATION

University of California, Davis, B.S. Computer Science

Expected Jun 2028

Coursework: Data Structures & Algorithms (C++), Object-Oriented Programming (C++), Assembly Programming

EXPERIENCE

AI Intern, Demolish Foods

Aug 2026 – Sep 2026

- Built an internal search and Q&A tool end to end (FastAPI, PostgreSQL, React/TypeScript) that lets food scientists find past experiments and trace every answer back to its source document; demoed it live to the R&D team
- Wrote the ingestion pipeline that turns a shared-drive archive into structured experiment records, with human review before a record goes live
- Used deterministic extraction for facts and an LLM only for Q&A, with source citations and flags when sources conflict or data is missing
- Added Google SSO (OIDC) and role-based access, and covered the system with pytest and Playwright tests

Undergraduate Research Assistant, Z-Lab, UC Davis

May 2026 – Present

- Built a sleep-staging pipeline for whole-night intracranial EEG (14 patients, 32K+ scored epochs, 100+ GB)
- Trained a gradient-boosting sleep-stage classifier that reached Cohen's κ 0.56 vs. 0.39 for the published SleepSEEG algorithm (3-class Wake/NREM/REM, leave-one-subject-out CV)
- Traced anomalous benchmark results to a feature-definition error and a ground-truth label mismatch, and corrected the evaluation

Founder & Lead Engineer, Efferent Systems (PyBCI)

Jun 2025 – Present

- Lead a 5-person team building PyBCI, a desktop platform (Python, PyQt5, FastAPI) for EEG acquisition, preprocessing, visualization, and ML model training; shipped v1.0, now used by UC Davis Neurotech members
- Set up GitHub Actions CI/CD (tests, release builds, SDK publishing) around a ~940-test pytest suite
- Built SandBox, a headless harness that replays PyBCI pipelines on synthetic EEG in CI to catch regressions without hardware

PROJECTS

AutoBuild: Multi-Agent Coding Orchestrator

Python, FastAPI, SQLite, Podman

- Built an orchestrator that routes feature requests through PM, tech-lead, engineer, and QA agents (Claude Code, Codex, DeepSeek), with human approval gates and an audit log of every agent action
- Ran generated code in network-isolated rootless Podman sandboxes, and made CI reproducible by recording LLM outputs and replaying them deterministically

Meridian: Semantic Graph Platform (open source, live demo)

TypeScript, React, SQLite

- Built a platform that parses Markdown notes and code repositories into one semantic graph, then diffs, validates, and visualizes it in a React studio; new source types plug in through a versioned plugin API

Second Brain: Personal AI Agent

Python, Claude Code, SQLite, discord.py

- Built an always-on agent that indexes email, chat, and calendar data into a SQLite full-text store and triages it into tasks, events, and a daily brief
- Exposed it through a Discord bot that runs headless Claude Code with read-only tools, locked to a single-user allowlist

LEADERSHIP

Board Member & Projects Division Lead, UC Davis Neurotech

2025 – Present

- Run technical design reviews of member BCI projects (paradigms, features, overfitting); lead NeuralVLA, an EEG- and vision-language-action-controlled robotic arm, with a MuJoCo 7-DOF simulation
- Assembled and led a 5-person team at Global NeuroHack 2026, building an EEG + fNIRS speech decoder with contrastive cross-modal embeddings (PyTorch)

SKILLS

Languages: Python, C++, TypeScript, SQL

ML / Data: PyTorch, scikit-learn, TensorFlow, NumPy, pandas, SciPy, MNE (EEG), LLM agents, RAG

Backend / Frontend: FastAPI, PostgreSQL, SQLite, React, PyQt5

Tools: Git, GitHub Actions, Docker, Podman, Linux, Google Cloud, pytest, Playwright